

CODEFEST 7:

1) a) Compute dense mvm (14N) 4

For matrix-vector multiply.

$$\begin{aligned}\#FLOPs &= 2 \times N^2 \\ &= 2 \times (512)^2 \\ &= 524,288 \text{ FLOPs}\end{aligned}$$

b) Compute dense memory bytes

$$\begin{aligned}\text{Dense memory bytes} &= 4 \times \text{FLOPs for } N^2 \\ &= 4N^2 \\ &= 4 \times (512)^2 \\ &= 1,048,576 \text{ bytes}\end{aligned}$$

c) Compute sparse computation using CSR (as fn of 's')

$$\begin{aligned}\#FLOPs &= (\text{number of non-zero elements}) \times (\text{total elements}) \times 2 \\ &= (1-s) \times N^2 \times 2 \quad ; \text{ the 2 is FLOPs for 1 add and 1 mul. } \end{aligned}$$

$$= 2(1-s) \times N^2$$

2) Compute sparse bytes (as fn of 's')

$$\begin{aligned}\text{Sparse memory bytes} &= 3 \text{ arrays} \\ &= \text{Values array} + \text{Col-idn array} + \text{Row-ptr array}.\end{aligned}$$

$$\begin{aligned}\text{Values array} &= \text{FP32 per non-zero element (hence, multiply by 4)} \\ &= 4 \times (1-s) N^2 \text{ bytes}\end{aligned}$$

$$\begin{aligned}\text{Col-idn array} &= \text{one INT32 per non-zero column} \\ &= 4 \times (1-s) N^2 \text{ bytes}\end{aligned}$$

Row-pta array = one INT32 per row pointer.
 $= 4(N+1)$ bytes

The +1 is added as the algorithm takes $(N+1)$ and stores 1 more element in the row-pta array.

Total Memory for sparse = Values array + (Col-idx + Row-pta)
 $= 8(1-s)N^2 + 4(N+1)$ bytes.

For large values of N , $4(N+1)$ term can be ignored.

Total memory for sparse = $8(1-s)N^2$

For $N=512$:

$$\begin{aligned} \text{Total memory} &= 8(1-s)(512)^2 + 4(513) \\ &= 2,097,152(1-s) + 2052 \text{ bytes.} \end{aligned}$$

2) Speedup for sparse vs dense MVM

$$\text{Speedup} = \frac{\text{Dense MVM FLOPs}}{\text{Sparse CSR FLOPs}}$$

$$= \frac{2N^2}{2(1-s)N^2}$$

$$\text{Speedup} = \frac{1}{1-s}$$

Computing 's' for speedup = 2x

$$\text{Speedup} = \frac{1}{1-s}$$

$$2 = \frac{1}{1-s}$$

$$1-s = 0.5$$

$$s = 0.5$$

If matrix is 50% sparse, speedup will be 2x.

3) Compute 's' when sparse memory = dense memory

When both are equal

$$8(1-s)N^2 + 4(N+1) = 4(N^2)$$

For 'N' very large, $4(N+1)$ can be ignored.

$$2 \cdot 8(1-s)N^2 = 4N^2$$

$$1-s = 0.5$$

$$s = 0.5$$

Above 50% sparsity, the CSR computation uses less memory.

4) Compute execution time speedup for sparse vs dense computation.

a) Dense memory = $4N^2 = 1,048,576$ bytes

$$\text{Sparse memory} = 8(1-s)N^2 + 4(N+1)$$

$$'s' = 0.9 \text{ (given)}$$

$$\text{Sparse memory} = 8(1 - 0.9)(512)^2 + 4(513)$$

$$= 209,715.2 + 2052$$

$$= 211,767.2 \text{ bytes}$$

b) Execution time:

$$\text{Dense time} = \frac{\text{bytes}}{\text{bandwidth}}$$

$$= \frac{1,048,576}{320 \times 10^9}$$

$$= 3.28 \times 10^{-6} \text{ seconds}$$

$$\text{Sparse time} = \frac{211,767}{320 \times 10^9}$$

$$= 6.618 \times 10^{-7} \text{ seconds}$$

$$\text{c) Speedup of execution time} = \frac{\text{Dense time}}{\text{Sparse time}}$$

$$= \frac{3.28 \times 10^{-6}}{6.618 \times 10^{-7}}$$

$$= 4.95$$

For memory-bound case, speedup is almost 5x